Vector Integration

1

Define - Line integral, Surface integral, Volume integral.

Line integral

Let $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$, where $\bar{r}$ is the position vector of (x,y,z) on a curve C joining points P_1 and P_2. Let $\bar{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$ be a continuous vector function along C. Then the line integral of the tangential component of $\bar{A}$ along C from P_1 to P_2 is defined by

$$\int_{P_1}^{P_2} \bar{A}.d\bar{r} = \int_C \bar{A}.d\bar{r} = \int_C A_1dx + A_2dy + A_3dz$$

Surface integral

Let S be a two-sided surface. Consider a unit normal $\bar{n}$ to any point of the positive side of S called the positive or outward drawn unit normal. Associate with the differential of surface area dS a vector $d\bar{S}$ whose magnitude is dS and whose direction is that of $\bar{n}$. Then $d\bar{S} = \bar{n}\,dS$

Then the integral

$$\iint_S \bar{A}.d\bar{S} = \iint_S \bar{A}.\bar{n}\,dS$$

Volume integral

Consider a closed surface in space enclosing a volume V. Then the integral

$$\iiint_V \bar{A}\, dV \quad \text{or} \quad \iiint_V \phi\, dV$$

is called volume integral.

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1. If $\bar{A} = (3x^2+6y)\hat{i} - 14yz\,\hat{j} + 20xz^2\hat{k}$, evaluate $\int_c \bar{A}.d\bar{r}$ from $(0,0,0)$ to $(1,1,1)$ along the following paths c

a) $x=t, y=t^2, z=t^3$

b) the straight lines from $(0,0,0)$ to $(1,0,0)$ then to $(1,1,0)$ and then to $(1,1,1)$

Solution:- Given that

$$\bar{A} = (3x^2+6y)\hat{i} - 14yz\,\hat{j} + 20xz^2\hat{k}$$

Let $\bar{r} = x\hat{i}+y\hat{j}+z\hat{k}$ be the position vector

so that $d\bar{r} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$.

Now, $\int_c \bar{A}.d\bar{r} = \int_c \{(3x^2+6y)\hat{i} - 14yz\,\hat{j} + 20xz^2\hat{k}\}.(dx\,\hat{i}+dy\,\hat{j}+dz\,\hat{k})$

$= \int_c (3x^2+6y)x - 14y^2z + 20xz^3$

$= \int_c 3x^3 + 6xy$

Now, $\int_C \bar{A}.d\bar{r} = \int_C \{(3x^2+6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}\}.(dx\hat{i}+dy\hat{j}+dz\hat{k})$

$= \int_C (3x^2+6y)\,dx - 14yz\,dy + 20xz^2\,dz \quad - - - - (1)$

a) If $x=t$, $y=t^2$ and $z=t^3$ when x goes to 0 to 1 then t goes to 0 to 1. Also it is true for y and z.

Therefore

$\int_C \bar{A}.d\bar{r} = \int_0^1 (3t^2+6t^2)\,d(t) - 14.t^2.t^3\,d(t^2) + 20.t.(t^3)^2\,d(t^3)$

$= \int_0^1 9t^2\,dt - 14t^5.2t\,dt + 20t^7.3t^2\,dt$

$= \int_0^1 9t^2\,dt - 28t^6\,dt + 60t^9\,dt$

$= \left[9.\frac{t^3}{3} - 28\,\frac{t^7}{7} + 60\,\frac{t^{10}}{10}\right]_0^1$

$= \left[3t^3 - 4t^7 + 6t^{10}\right]_0^1$

$= 5$

b) Along the straight line from (0,0,0) to (1,0,0) i.e. x varies from 0 to 1 but $y=0, z=0$ so that $dy=0$, $dz=0$, ~~we get from (1)~~, then the integral over this part of the path is

$\int_{x=0}^1 (3x^2+6.0)\,dx - 14.0 + 20.0$

$= \int_0^1 3x^2\,dx$

$$= \left[3.\frac{x^3}{3}\right]_0^1 = \left[x^3\right]_0^1 = 1$$

Along the straight line from (1,0,0) to (1,1,0) i.e. $x=1, z=0$ so that $dx=0, dz=0$ while y varies from 0 to 1. Then the integral over this part of the path is

$$\int_{y=0}^{1} (3.1+6y)\,0 - 14y.0\,dy + 20.1.0$$

$$= 0$$

Along the straight line from (1,1,0) to (1,1,1) $x=1, y=1$ so that $dx=dy=0$ while z varies from 0 to 1. Then the integral over this part of the path is

$$\int_{z=0}^{1} (3.1+6.1)\,0 - 14.1.z.0 + 20.1.z^2\,dz$$

$$= \int_0^1 20z^2\,dz = \left[\frac{20}{3}z^3\right]_0^1 = \frac{20}{3}$$

Therefore, adding (2), (3), (4) we get

$$\int_C \bar{A}.d\bar{r} = 1+0+\frac{20}{3} = \frac{23}{3}$$

——— x ———

c) the straight line joining (0,0,0) and (1,1,1).

Solution:- The straight line joining (0,0,0) and (1,1,1) is given in parametric form by $x=t, y=t, z=t$.

Then from (1), we get

$$\int_C \bar{A}.d\bar{r} = \int_{t=0}^{1} (3t^2+6t)dt - 14t^2dt + 20t^3dt$$

$$= \int_0^1 \left(3t^2+6t-14t^2+20t^3\right)dt$$

$$= \int_0^1 \left(6t-11t^2+20t^3\right)dt$$

$$= \left[\frac{6}{2}t^2 - \frac{11}{3}t^3 + \frac{20}{4}t^4\right]_0^1$$

$$= \left[3t^2 - \frac{11}{3}t^3 + 5t^4\right]_0^1$$

$$= 3 - \frac{11}{3} + 5$$

$$= \frac{13}{3}$$

———— X ————

6

2. Find the total work done in moving a particle in a force field given by $\bar{F} = 3xy\hat{i} - 5z\hat{j} + 10x\hat{k}$ along the curve $x = t^2+1$, $y = 2t^2$, $z = t^3$ from $t=1$ to $t=2$.

Solution:- Given that

$$\bar{F} = 3xy\hat{i} - 5z\hat{j} + 10x\hat{k}$$

Let $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$ so that $d\bar{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

Now $\bar{F}.d\bar{r} = (3xy\hat{i} - 5z\hat{j} + 10x\hat{k}).(dx\hat{i} + dy\hat{j} + dz\hat{k})$

$$= 3xy\,dx - 5z\,dy + 10x\,dz$$

Therefore, the total work done along the curve $x = t^2+1$, $y = 2t^2$, $z = t^3$ from $t=1$ to $t=2$ is

$$= \int_{t=1}^{2} 3.(t^2+1).2t^2\,d(t^2+1) - 5.t^3\,d(2t^2) + 10.(t^2+1)\,d(t^3)$$

$$= \int_1^2 6t^2(t^2+1).2t\,dt - 5t^3.4t\,dt + 10(t^2+1).3t^2\,dt$$

$$= \int_1^2 12t^3(t^2+1)\,dt - 20t^4\,dt + 30t^2(t^2+1)\,dt$$

$$= \int_1^2 (12t^5 + 12t^3 - 20t^4 + 30t^4 + 30t^2)\,dt$$

$$= \int_1^2 (12t^5 + 10t^4 + 12t^3 + 30t^2)\,dt$$

$$= \left[\frac{12}{6}t^6 + \frac{10}{5}t^5 + \frac{12}{4}t^4 + \frac{30}{3}t^3\right]_1^2$$

$$= \left[2t^6 + 2t^5 + 3t^4 + 10t^3\right]_1^2$$

$$= 303.$$

—— X ——

(3) If $\bar{F} = 3xy\,\hat{i} - y^2\,\hat{j}$ evaluate $\int_C \bar{F}.d\bar{r}$ where C is the curve in the xy plane $y = 2x^2$ from (0,0) to (1,2)

Solution:- Given that

$$\bar{F} = 3xy\,\hat{i} - y^2\,\hat{j}$$

In the xy plane $z = 0$ so that

$$\bar{r} = x\hat{i} + y\hat{j} \quad \text{then} \quad d\bar{r} = dx\,\hat{i} + dy\,\hat{j}$$

Now $\bar{F}.d\bar{r} = (3xy\,\hat{i} - y^2\,\hat{j}).(dx\,\hat{i} + dy\,\hat{j})$

$$= 3xy\,dx - y^2\,dy$$

Therefore, along point (0,0) to (1,2) and $y = 2x^2$ where x goes from 0 to 1, we get

$$\int_C \bar{F}.d\bar{r} = \int_{x=0}^{1} 3x.2x^2\,dx - (2x^2)^2\,d(2x^2)$$

$$= \int_0^1 6x^3\,dx - 4x^4.4x\,dx$$

$$= \int_0^1 \left(6x^3 - 16x^5\right) dx$$

$$= \left[\frac{6}{4}x^4 - \frac{16}{6}x^6\right]_0^1$$

$$= \frac{6}{4} - \frac{16}{6} \quad \frac{3}{2} - \frac{8}{3}$$

$$= -\frac{7}{6}.$$

———— X ————

4. Find the work done in moving a particle once around a circle C in the xy plane, if the circle has centre at the origin and radius 3 and if the force field is given by

$$\bar{F} = (2x-y+z)\hat{i} + (x+y-z^2)\hat{j} + (3x-2y+4z)\hat{k}$$

Solution:- Given that

$$\bar{F} = (2x-y+z)\hat{i} + (x+y-z^2)\hat{j} + (3x-2y+4z)\hat{k}$$

In the xy plane $z=0$, so that $\bar{r} = x\hat{i} + y\hat{j}$

then $d\bar{r} = dx\hat{i} + dy\hat{j}$ and also

$$\bar{F} = (2x-y)\hat{i} + (x+y)\hat{j} + (3x-2y)\hat{k}$$

Now $\bar{F}.d\bar{r} = \{(2x-y)\hat{i} + (x+y)\hat{j} + (3x-2y)\hat{k}\}.(dx\hat{i} + dy\hat{j})$

$$= (2x-y)dx + (x+y)dy$$

Given that the circle has centre at the origin with radius 3. Then ~~the~~ its equation is

$$x^2+y^2=3^2 \quad - - - (1)$$

Let $x=3\cos\theta$, $y=3\sin\theta$ be the parametric equation of (1) and θ varies from 0 to 2π. So that the work done is

$$\int_C \bar{F}.d\bar{r} = \int_{\theta=0}^{2\pi} (2.3\cos\theta - 3\sin\theta)\, d(3\cos\theta) + (3\cos\theta + 3\sin\theta)\, d(3\sin\theta)$$

$$= \int_0^{2\pi} (6\cos\theta - 3\sin\theta)(-3\sin\theta)\, d\theta + (3\cos\theta + 3\sin\theta)(3\cos\theta)\, d\theta$$

$$= \int_0^{2\pi} (-18\sin\theta\cos\theta + 9\sin^2\theta)\, d\theta + (9\cos^2\theta + 9\sin\theta\cos\theta)\, d\theta$$

$$= \int_0^{2\pi} (-18\sin\theta\cos\theta + 9\sin^2\theta + 9\cos^2\theta + 9\sin\theta\cos\theta)\, d\theta$$

$$= \int_0^{2\pi} (9\sin^2\theta - 9\sin\theta\cos\theta + 9\cos^2\theta)\, d\theta$$

$$= \int_0^{2\pi} 9(\sin^2\theta+\cos^2\theta)\, d\theta - \int_0^{2\pi} 9\sin\theta\cos\theta\, d\theta$$

$$= \int_0^{2\pi} 9\, d\theta - \frac{9}{2}\int_0^{2\pi} 2\sin\theta\cos\theta\, d\theta$$

$$= 9[\theta]_0^{2\pi} - \frac{9}{2}\int_0^{2\pi} \sin 2\theta \, d\theta$$

$$= 9.2\pi - \frac{9}{2}\left[\frac{-\cos 2\theta}{2}\right]_0^{2\pi}$$

$$= 18\pi + \frac{9}{2}\left[\frac{\cos 2\theta}{2}\right]_0^{2\pi}$$

$$= 18\pi + \frac{9}{2}\left[\frac{\cos 2.2\pi}{2} - \frac{\cos 0}{2}\right]$$

$$= 18\pi + \frac{9}{2}\left(\frac{1}{2} - \frac{1}{2}\right)$$

$$= 18\pi.$$

In moving C, we have choosen the counterclockwise direction i.e. positive direction. If the particle moves clockwise direction i.e. negative direction then the value of the integral would be -18π.

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. ✱ (a) If $\bar{F}$ is a conservative field, prove that curl $\bar{F} = \bar{\nabla} \times \bar{F} = 0$ (i.e. $\bar{F}$ is irrotational)

(b) If $\bar{\nabla} \times \bar{F} = 0$ (i.e. $\bar{F}$ is irrotational), prove that $\bar{F}$ is conservative.

Solution:- We know that, if $\bar{F}$ is a conservative field, then $\bar{F} = \bar{\nabla}\phi$.

Thus curl $\bar{F} = \bar{\nabla} \times \bar{F}$

$= \bar{\nabla} \times \bar{\nabla}\phi$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix}$$

$$= \left(\frac{\partial}{\partial y}\frac{\partial \phi}{\partial z} - \frac{\partial}{\partial z}\frac{\partial \phi}{\partial y}\right)\hat{i} + \left(\frac{\partial}{\partial z}\frac{\partial \phi}{\partial x} - \frac{\partial}{\partial x}\frac{\partial \phi}{\partial z}\right)\hat{j} + \left(\frac{\partial}{\partial x}\frac{\partial \phi}{\partial y} - \frac{\partial}{\partial y}\frac{\partial \phi}{\partial x}\right)\hat{k}$$

$= 0$

(Proved)

(b) Let us consider $\bar{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$

If $\bar{\nabla} \times \bar{F} = 0$, then

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = 0$$

$$\Rightarrow \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\hat{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right)\hat{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\hat{k} = 0$$

Thus we get,

$$\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} = 0 \;,\; \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} = 0 \;,\; \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = 0$$

$$\text{or } \frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z} \;,\; \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x} \;,\; \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y} \quad \text{---} \quad (*)$$

To prove that F is conservative, we have to show that $\bar{F} = \bar{\nabla}\phi$.

Now the work done in moving a particle from (x_1, y_1, z_1) to (x, y, z) in the force field $\bar{F}$ is

$$\int_C F_1(x,y,z)\,dx + F_2(x,y,z)\,dy + F_3(x,y,z)\,dz$$

where C is a path joining (x_1, y_1, z_1) and (x, y, z).

Let us choose a particular path the straight line segments from (x_1, y_1, z_1) to (x, y_1, z_1) to (x, y, z_1) to (x, y, z) and say $\phi(x, y, z)$ the work done along this particular path. ~~Thus~~ Then

$$\phi(x,y,z) = \int_{x_1}^{x} F_1(x, y_1, z_1)\, dx + \int_{y_1}^{y} F_2(x, y, z_1)\, dy + \int_{z_1}^{z} F_3(x, y, z)\, dz$$

Now, $\frac{\partial \phi}{\partial z} = \frac{\partial}{\partial z}\left[\int_{z_1}^{z} f_3(x, y, z)\, dz\right]$

$$= F_3(x, y, z)$$

$$\frac{\partial \phi}{\partial y} = \frac{\partial}{\partial y}\left[\int_{y_1}^{y} F_2(x, y, z_1)\, dy\right] + \int_{z_1}^{z} \frac{\partial f_3}{\partial y}(x, y, z)\, dz$$

$$= F_2(x, y, z_1) + \int_{z_1}^{z} \frac{\partial F_2}{\partial y \partial z}(x, y, z)\, dz \quad \left[\because \frac{\partial f_3}{\partial y} = \frac{\partial f_2}{\partial z}\right]$$

$$= F_2(x, y, z_1) + \left[F_2(x, y, z)\right]_{z_1}^{z}$$

$$= F_2(x, y, z_1) + F_2(x, y, z) - F_2(x, y, z_1)$$

$$= F_2(x, y, z)$$

$\frac{\partial \phi}{\partial x} =$

$$\frac{\partial \phi}{\partial x} = \frac{\partial}{\partial x}\int_{x_1}^{x} F_1(x, y_1, z_1)\,dx + \int_{y_1}^{y} \frac{\partial F_2}{\partial x}(x, y, z_1)\,dy$$

$$+ \int_{z_1}^{z} \frac{\partial F_3}{\partial x}(x, y, z)\,dz$$

$$\left[\because \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y} \quad \frac{\partial F_3}{\partial x} = \frac{\partial F_1}{\partial z}\right]$$

$$= F_1(x, y_1, z_1) + \int_{y_1}^{y} \frac{\partial F_1}{\partial y}(x, y, z_1)\,dy$$

$$+ \int_{z_1}^{z} \frac{\partial F_1}{\partial z}(x, y, z)\,dz$$

$$= F_1(x, y_1, z_1) + \left[F_1(x, y, z_1)\right]_{y_1}^{y} + \left[F_1(x, y, z)\right]_{z_1}^{z}$$

$$= F_1(x, y_1, z_1) + F_1(x, y, z_1) - F_1(x, y_1, z_1)$$

$$+ F_1(x, y, z) - F_1(x, y, z_1)$$

$$= F_1(x, y, z)$$

Therefore,

$$F = F_1(x, y, z)\,\hat{i} + F_2(x, y, z)\,\hat{j} + F_3(x, y, z)\,\hat{k}$$

$$= \frac{\partial \phi}{\partial x}\,\hat{i} + \frac{\partial \phi}{\partial y}\,\hat{j} + \frac{\partial \phi}{\partial z}\,\hat{k}$$

$$= \bar{\nabla}\phi.$$

Hence, $\bar{F}$ is conservative.

——— X ———

6. ## (a) Show that $\bar{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$ is a conservative force field

(b) Find the scalar potential

(c) Find the work done in moving an object in this field from $(1,-2,1)$ to $(3,1,4)$

Solution:-

(a) We know that, a necessary and sufficient condition that a force will be conservative is that $\text{curl}\,\bar{F} = \bar{\nabla} \times \bar{F} = 0$.

Given that, $\bar{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$.

Now, $\text{curl}\,\bar{F} = \bar{\nabla} \times \bar{F}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy+z^3 & x^2 & 3xz^2 \end{vmatrix}$$

$= 0$

Thus $\bar{F}$ is a conservative force field.

(b) Since $\bar{F}$ is conservative, $\int_C \bar{F}.d\bar{r}$ is independent of the path C joining (x_1, y_1, z_1) and (x, y, z).

Therefore, the scalar potential is

$$\phi(x,y,z) = \int_{x_1}^{x} (2xy_1 + z_1^3)\,dx + \int_{y_1}^{y} x^2\,dy + \int_{z_1}^{z} 3xz^2\,dz$$

$$= \left[x^2y_1 + xz_1^3\right]_{x_1}^{x} + \left[x^2y\right]_{y_1}^{y} + \left[xz^3\right]_{z_1}^{z}$$

$$= x^2y_1 + xz_1^3 - x_1^2y_1 - x_1z_1^3 + x^2y - x^2y_1 + xz^3 - xz_1^3$$

$$= x^2y + xz^3 - x_1^2y_1 - x_1z_1^3$$

$$= x^2y + xz^3 + \text{constant}.$$

(c) Given $\bar{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$.

The work done in moving an object is

$$= \int_{P_1}^{P_2} \bar{F}.d\bar{r}$$

$$= \int_{P_1}^{P_2} (2xy + z^3)dx + x^2\,dy + z^2\,dz\; 3xz^2\,dz$$

$$= \int_{P_1}^{P_2} d\left(x^2y + xz^3\right)$$

$$= \left[x^2y + xz^3\right]_{P_1}^{P_2}$$

Therefore, the work done from $(1,-2,1)$ to $(3,1,4)$ is

~~Work done~~ $= \left[x^2y + xz^3\right]_{(1,-2,1)}^{(3,1,4)}$

$= 202.$

——— X ———

7. ## If $\phi = 2xyz^2$, $\bar{F} = xy\hat{i} - z\hat{j} + x^2\hat{k}$ and C is the curve $x = t^2, y = 2t, z = t^3$ from $t=0$ to $t=1$, evaluate the line integrals (a) $\int_C \phi \, d\bar{r}$ (b) $\int_C \bar{F} \times d\bar{r}$.

Solution:-

(a) Given, $\phi = 2xyz^2$. We know, $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$

Along C i.e. $x = t^2, y = 2t, z = t^3$, we have

$$\phi = 2xyz^2 = 2 \cdot t^2 \cdot 2t \cdot (t^3)^2$$

$$= 4t^9$$

$$\bar{r} = x\hat{i} + y\hat{j} + z\hat{k} = t^2\hat{i} + 2t\hat{j} + t^3\hat{k}$$

$$\therefore d\bar{r} = (2t\hat{i} + 2\hat{j} + 3t^2\hat{k})\,dt$$

Now, from $t=0$ to $t=1$,

$$\int_C \phi \, d\bar{r} = \int_{t=0}^{1} 4t^9 (2t\hat{i} + 2\hat{j} + 3t^2\hat{k})\,dt$$

$$= \int_0^1 (8t^{10}\hat{i} + 8t^9\hat{j} + 12t^{11}\hat{k})\,dt$$

$$= \left[\frac{8}{11}t^{11}\hat{i} + \frac{8}{10}t^{10}\hat{j} + \frac{12}{12}t^{12}\hat{k}\right]_0^1$$

$$= \frac{8}{11}\hat{i} + \frac{4}{5}\hat{j} + \hat{k}$$

(b) Given, $\bar{F} = xy\hat{i} - z\hat{j} + x^2\hat{k}$. We have $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

Along C i.e. $x = t^2$, $y = 2t$, $z = t^3$, we have

$$\bar{F} = xy\hat{i} - z\hat{j} + x^2\hat{k}$$

$$= t^2 \cdot 2t\,\hat{i} - t^3\hat{j} + (t^2)^2\hat{k}$$

$$= 2t^3\hat{i} - t^3\hat{j} + t^4\hat{k}$$

and $d\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$= t^2\hat{i} + 2t\hat{j} + t^3\hat{k}$$

$$\therefore d\bar{r} = \left(2t\hat{i} + 2\hat{j} + 3t^2\hat{k}\right)dt$$

Then $\bar{F} \times d\bar{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2t^3 & -t^3 & t^4 \\ 2t & 2 & 3t^2 \end{vmatrix}$

dt

$$= \left\{\hat{i}\left(-3t^5 - 2t^4\right) + \hat{j}\left(2t^5 - 6t^5\right) + \hat{k}\left(4t^3 + 2t^4\right)\right\}$$

$$= \left\{\left(-3t^5 - 2t^4\right)\hat{i} - 4t^5\hat{j} + \left(4t^3 + 2t^4\right)\hat{k}\right\}dt$$

Hence, from $t=0$ to $t=1$,

$$\int_C \bar{F} \times d\bar{r} = \int_0^1 \left\{ \left(-3t^5 - 2t^4\right)\hat{i} - 4t^5\,\hat{j} + \left(4t^3 + 2t^4\right)\hat{k} \right\} dt .$$

$$= \left[\left(-\frac{3}{6}t^6 - \frac{2}{5}t^5\right)\hat{i} - \frac{4}{6}t^6\,\hat{j} + \left(\frac{4}{4}t^4 + \frac{2}{5}t^5\right)\hat{k} \right]_0^1$$

$$= -\frac{27}{30}\hat{i} - \frac{4}{6}\hat{j} + \frac{7}{5}\hat{k}$$

$$= -\frac{9}{10}\hat{i} - \frac{2}{3}\hat{j} + \frac{7}{5}\hat{k}$$

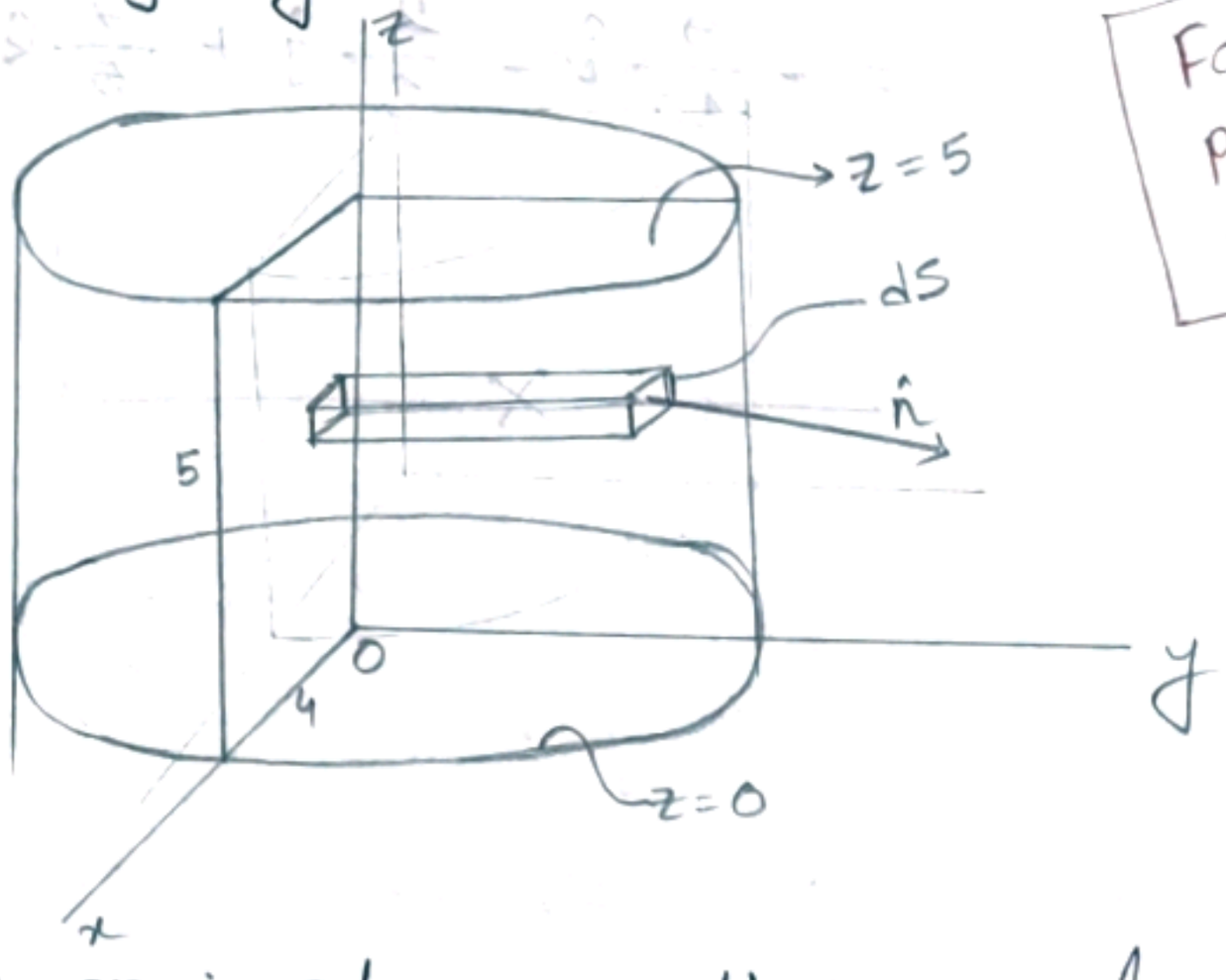
———— X ————

Surface Integral

3. ** Evaluate $\iint_S \bar{A}\cdot\bar{n}\,dS$, where $\bar{A} = z\hat{i} + x\hat{j} - 3y^2z\hat{k}$ and S is the surface of the cylinder $x^2+y^2=16$ included in the first octant between $z=0$ and $z=5$.

Solution:- Given that

$$\bar{A} = z\hat{i} + x\hat{j} - 3y^2z\hat{k}$$



For better picture see from book

At first we project S on the xz plane as in the figure above and called the projection R. Then

$$\iint_S \bar{A}\cdot\bar{n}\,dS = \iint_R \bar{A}\cdot\hat{n}\,\frac{dx\,dz}{|\hat{n}\cdot\hat{j}|}$$

[$\hat{n}$ = unit normal]

Also given $x^2+y^2=16$. Then a normal to $x^2+y^2=16$ is $\bar{\nabla}(x^2+y^2) = \frac{\partial}{\partial x}(x^2+y^2)\hat{i} + \frac{\partial}{\partial y}(x^2+y^2)\hat{j} + \frac{\partial}{\partial z}(x^2+y^2)\hat{k}$

$= 2x\hat{i} + 2y\hat{j}$

Thus the unit normal to S is

$$\hat{n} = \frac{2x\hat{i}+2y\hat{j}}{\sqrt{(2x)^2+(2y)^2}}$$

$$= \frac{2x\hat{i}+2y\hat{j}}{\sqrt{4x^2+4y^2}} = \frac{2x\hat{i}+2y\hat{j}}{\sqrt{4(x^2+y^2)}} = \frac{2x\hat{i}+2y\hat{j}}{\sqrt{4.16}}$$

$$= \frac{2x\hat{i}+2y\hat{j}}{8} = \frac{x\hat{i}+y\hat{j}}{4} \quad [\because x^2+y^2=16]$$

Now, $\bar{A}\cdot\hat{n} = (z\hat{i}+x\hat{j}-3y^2z\hat{k})\cdot\left(\frac{x\hat{i}+y\hat{j}}{4}\right)$

$$= \frac{1}{4}(xz+xy)$$

$$\hat{n}\cdot\hat{j} = \frac{x\hat{i}+y\hat{j}}{4}\cdot\hat{j} = \frac{y}{4}$$

Then the surface integral equals

$$\iint_S \bar{A}\cdot\hat{n}\,dS = \iint_R \bar{A}\cdot\hat{n}\,\frac{dx\,dz}{|\hat{n}\cdot\hat{j}|}$$

$$= \int_{z=0}^{5}\int_{x=0}^{4} \frac{xz+xy}{y}\,dx\,dz$$

$$= \int_{z=0}^{5}\int_{x=0}^{4} \left(\frac{xz}{y}+x\right)dx\,dz$$

$$= \int_{z=0}^{5}\int_{x=0}^{4} \left(\frac{xz}{\sqrt{16-x^2}}+x\right)dx\,dz$$

$$= \int_{z=0}^{5} \left[\int_{x=0}^{4} \left(\frac{xz}{\sqrt{16-x^2}} + x \right) dx \right] dz$$

$$= \int_{z=0}^{5} \left[z \int_{x=0}^{4} \frac{x}{\sqrt{16-x^2}} dx + \int_{x=0}^{4} x \, dx \right] dz$$

Let us put $16 - x^2 = z$, then $-2x\,dx = dz$

$$\Rightarrow x\,dx = -\frac{1}{2} dz.$$

For limit, when $x=0$, then $z=16$

when $x=4$, then $z=0$

So that, we get

$$-\frac{1}{2} \int_{z=16}^{0} \frac{dz}{\sqrt{z}} = -\frac{1}{2} 2 \left[\sqrt{z} \right]_{16}^{0} = -\left(0 - \sqrt{16}\right) = 4$$

and $$\int_0^4 x\,dx = \left[\frac{x^2}{2} \right]_0^4 = \frac{16}{2} = 8.$$

Therefore,

$$\iint_S \bar{A} \cdot \hat{n}\, dS = \int_{z=0}^{5} (4z + 8)\, dz$$

$$= \left[4 \cdot \frac{z^2}{2} + 8z \right]_0^5$$

$$= \left[2z^2 + 8z \right]_0^5$$

$$= 90.$$

——— x ———

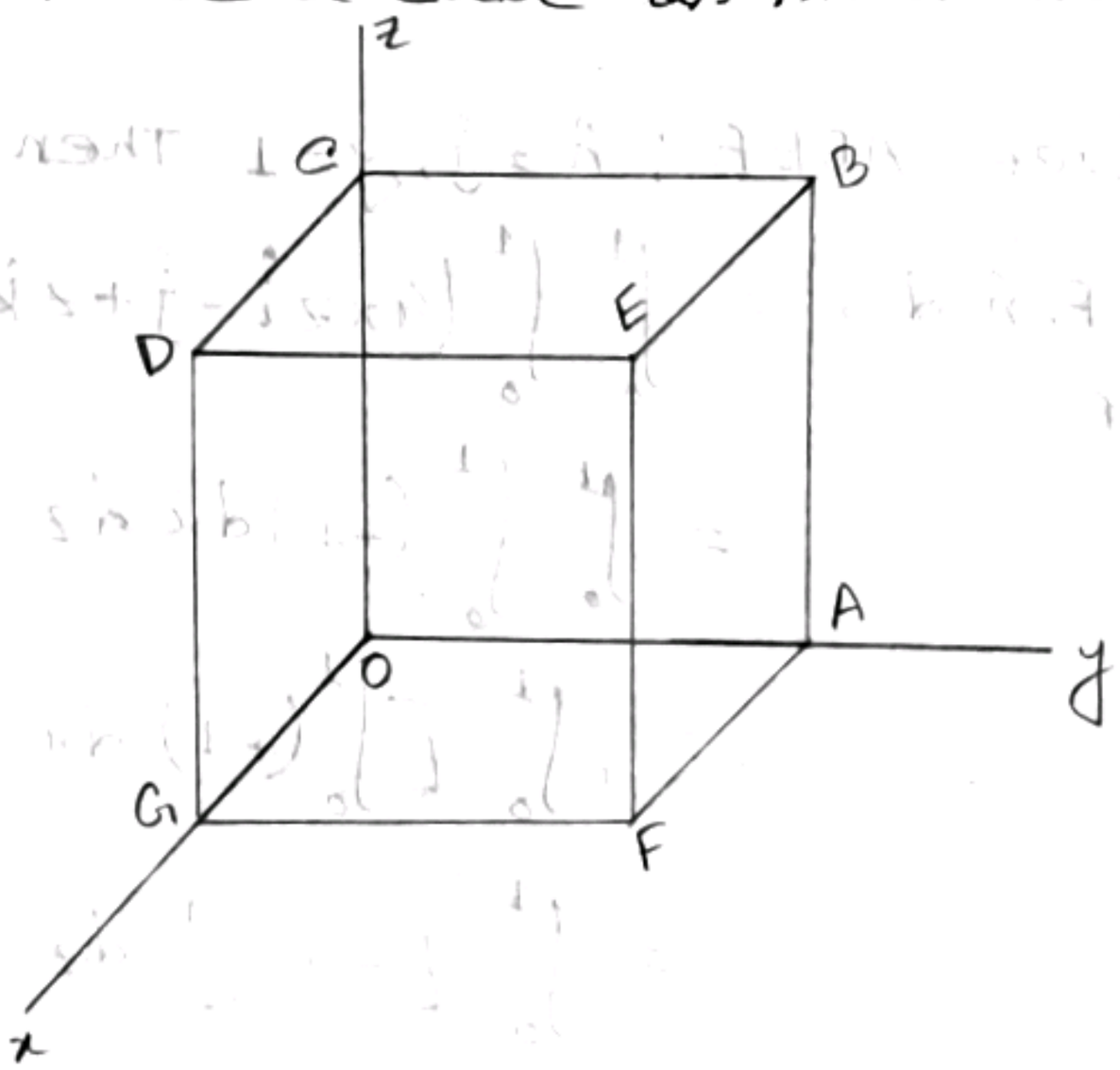
9. ** ASY

If $\bar{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$, evaluate $\iint_S \bar{F}.\hat{n}\, ds$ where S is the surface of the cube bounded by $x=0, x=1, y=0, y=1, z=0, z=1$

Solution :- Given that

$$\bar{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}.$$

Let ABCDEFGO be a cube as shown in figure below.



For face DEFG; $\hat{n} = \hat{i}$, $x = 1$. Then

$$\iint_{DEFG} \bar{F}.\hat{n}\, ds = \int_0^1 \int_0^1 \left(4z\hat{i} - y^2\hat{j} + yz\hat{k}\right).\hat{i}\; dy\, dz$$

Since $x=1, dx=0$ in $\bar{F}$ = given, Putting $x=1$. Then

$$= \int_0^1 \int_0^1 4z\, dy\, dz$$

$$= \int_0^1 z\left(\int_0^1 4z\, dy\right) dz$$

$$= \int_0^1 z\left[4y\right]_0^1 dz$$

z^4

$$= \int_0^1 4z\,dz = \left[4\frac{z^2}{2}\right]_0^1 = 2$$

For face ABCO; $\hat{n} = -\hat{i}$, $x = 0$. Then

$$\iint_{ABCO} \bar{F}.\hat{n}\,dS = \int_0^1 \int_0^1 \left(-y^2\hat{j} + yz\hat{k}\right)\cdot(-\hat{i})\,dy\,dz$$

$$= \int_0^1 \int_0^1 0$$

For face ABEF; $\hat{n} = \hat{j}$, $y = 1$. Then

$$\iint_{ABEF} \bar{F}.\hat{n}\,dS = \int_0^1 \int_0^1 \left(4xz\hat{i} - \hat{j} + z\hat{k}\right)\cdot\hat{j}\,dx\,dz$$

$$= \int_0^1 \int_0^1 (-1)\,dx\,dz$$

$$= \int_0^1 \left[\int_0^1 (-1)\,dx\right] dz$$

$$= \int_0^1 [-x]_0^1\,dz$$

$$= \int_0^1 -1\,dz$$

$$= [-z]_0^1$$

$$= -1.$$

For face ~~OGDC~~ OGDC; $\hat{n} = -\hat{j}$, $y = 0$. Then

$$\iint_{OGDC} \bar{F}.\hat{n}\,dS = \int_0^1 \int_0^1 (4xz\hat{i}).(-\hat{j})\,dx\,dz$$

$$= 0$$

For face BCDE ; $\hat{n} = \hat{k}$, $z = 1$. Then

$$\iint_{BCDE} \bar{F} \cdot \hat{n}\, dS = \int_0^1 \int_0^1 (4x\hat{i} - y^2\hat{j} + y\hat{k}) \cdot \hat{k}\, dx\, dy$$

$$= \int_0^1 \int_0^1 y\, dx\, dy$$

$$= \int_0^1 \left(\int_0^1 y\, dx \right) dy$$

$$= \int_0^1 [xy]_0^1\, dy$$

$$= \int_0^1 y\, dy$$

$$= \left[\frac{y^2}{2} \right]_0^1 = \frac{1}{2}$$

For face AFGO ; $\hat{n} = -\hat{k}$, $z = 0$. Then

$$\iint_{AFGO} \bar{F} \cdot \hat{n}\, dS = \int_0^1 \int_0^1 (-y^2\hat{j}) \cdot (-\hat{k})\, dx\, dy$$

$$= 0$$

Therefore, adding all the faces we get

$$\iint_S \bar{F} \cdot \hat{n}\, dS = 2 + 0 + (-1) + 0 + \frac{1}{2} + 0$$

$$= 1 + \frac{1}{2}$$

$$= \frac{3}{2}.$$

——— X ———

10. # Home Work

If $\bar{A} = (2y+3)\hat{i} + xz\hat{j} + (yz-x)\hat{k}$, evaluate $\int_C \bar{A} \cdot d\bar{r}$ along the following paths C:

(a) $x = 2t^2$, $y = t$, $z = t^3$ from $t=0$ to $t=1$.

(b) the straight lines from $(0,0,0)$ to $(0,0,1)$, then to $(0,1,1)$ and then to $(2,1,1)$

(c) the straight line joining $(0,0,0)$ and $(2,1,1)$

Answer:- (a) $\frac{288}{35}$ (b) 10 (c) 8.

——x——

Conservative Field

If $\bar{F} = \bar{\nabla}\phi$, where ϕ is single-valued and has continuous partial derivatives, then $\bar{F}$ is called conservative.

If $\bar{F}$ is conservative then curl $\bar{F} = \nabla \times \bar{F} = 0$.
i.e. $\bar{F}$ is irrotational.

$f(x) = x^2$ → single valued function
$f(x) = x$ → many valued function